

Marine Weather Decision System for Local Fishermen Using C++: Wind Speed and Wave Height Analysis for Immediate Departure Decisions

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Abstract—Fishermen who launch from remote coastal sites often face rapidly changing marine weather while having no cellular coverage to check forecast applications. Manual judgment of sea conditions is error-prone and can lead to unsafe departures. This report describes the background and motivation of the problem, defines it formally, and proposes an offline, rule-based Marine Weather Decision System written in C++. The program accepts two inputs, wind speed in knots and wave height in feet, and applies a conditional-branching algorithm to classify conditions as Gale Warning, Small Craft Advisory, or Safe, returning an immediate safety recommendation. The thresholds are adapted from National Weather Service marine definitions. The implementation was compiled and tested on seven representative input cases, and all produced the expected classification.

Index Terms—marine weather, decision support system, fishermen safety, offline computing, C++, conditional branching

I. INTRODUCTION

A. Background and Motivation

Small-scale fishing depends directly on sea conditions.

Wind and waves can change quickly, and a boat that departs under calm-looking conditions may meet dangerous weather within hours. Local fishermen, who commonly operate small boats, are especially exposed because such vessels have limited tolerance for strong wind and high waves.

Modern weather applications could help, but they depend on internet or cellular connectivity. Many coastal launch sites have no signal at all, which makes phone-based weather applications unusable exactly where the departure decision is made. As a result, fishermen often fall back on manual judgment, experience, or word of mouth. These observations motivate a tool that is simple, instant, and fully offline. This report proposes such a tool, implemented in C++, that converts two measurable quantities, wind speed and wave height, into a clear sail or do-not-sail recommendation.

B. Scope and Organization

The report covers the problem description (Section II), the proposed solution including its algorithm and implementation (Section III), test results (Section IV), and limitations and future work (Section V). Section VI concludes.

II. PROBLEM DESCRIPTION

The problem addressed in this work consists of three related issues:

1) Sudden weather changes. Local fishermen face rapid changes in marine weather, which leaves little time to react once at sea.

2) Misjudgment from manual checking. Assessing wind and waves by eye or by informal sources can lead to misjudgments and unsafe sea journeys, since the thresholds at which small boats become endangered are not always obvious.

3) No connectivity at launch sites. Many coastal launch sites have no cellular signal, so phone weather applications cannot be used when the decision is needed.

Formally, given a wind speed w (knots) and a wave height h (feet) available at the time of departure, the system must output one of three status classes together with an actionable safety recommendation, without requiring any network connection.

III. PROPOSED SOLUTION

A. Overview

We propose an automated safety recommendation program that tells the user, immediately, whether to sail or not based on wind speed and wave height input. The program runs locally on a low-power device and needs no external data source at run time.

B. Why C++

C++ was chosen for three reasons that match the field constraints. First, a compiled program has no network dependency, so it works in coastal zones with no internet or cellular connectivity. Second, it is lightweight and can run directly on low-power devices [2]. Third, the calculation is instantaneous and does not require heavy runtime dependencies.

C. Decision Logic

The program has three stages: input parsing, conditional branching, and output. Wind speed (knots) and wave height (feet) are read as floating-point variables. An **if-else if-else** structure then classifies the conditions, as summarized in TABLE I and Fig. 1. The hazard criteria are adapted from the National Weather Service marine definitions [1].

TABLE I
HAZARD CLASSIFICATION RULES

Status	Condition	Recommendation
Gale Warning	$w \geq 34$ kn	Extreme hazard; do not sail
Small Craft Advisory	$w \geq 20$ kn or $h \geq 7$ ft	Hazardous for small boats
Safe	All other cases	Conditions favorable for sailing

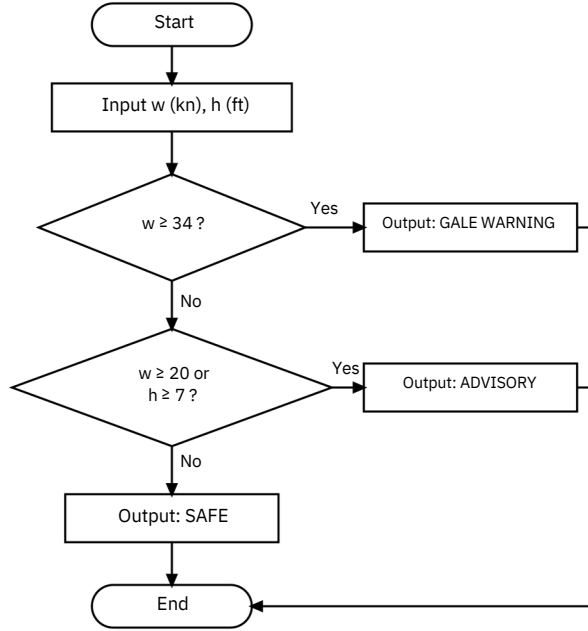


Fig. 1. Flowchart of the weather warning decision logic.

D. Implementation

The core program is a single C++ source file using only the standard input/output library. Listing 1 shows the decision section. Because the criteria are evaluated in order of severity, a Gale Warning always takes priority over an advisory.

```

if (wind_speed >= 34) {
    cout << "STATUS: GALE WARNING! "
          << "EXTREME HAZARD, DO NOT SAIL!";
}
else if (wind_speed >= 20 || wave_height >= 7.0) {
    cout << "STATUS: SMALL CRAFT ADVISORY! "
          << "HAZARDOUS FOR SMALL BOATS";
}
else {
    cout << "STATUS: SAFE. Conditions are "
          << "favorable for sailing";
}
  
```

Listing 1. Core decision logic in C++.

IV. RESULTS AND DISCUSSION

The program was compiled with g++ and executed on seven input cases chosen to cover each class and the boundary values. TABLE II lists the inputs and the observed output; all results match the rules in TABLE I.

TABLE II
TEST CASES AND OBSERVED OUTPUT

Wind (kn)	Wave (ft)	Output status
10	3	Safe
19.9	6.9	Safe
20	3	Small Craft Advisory
15	7	Small Craft Advisory
25	4	Small Craft Advisory
34	2	Gale Warning
37	7	Gale Warning

The boundary cases confirm the inclusive comparison operators: a wind of exactly 20 knots or a wave of exactly 7 feet triggers an advisory, while 19.9 knots with 6.9 feet remains safe. The input 37 knots with 7 feet returns a Gale Warning, showing that wind speed above the gale threshold overrides the wave condition.

The system gives a result as soon as the two values are entered, requires no network, and its logic is simple enough for non-technical users to understand. These properties directly address the three problems in Section II.

V. LIMITATIONS AND FUTURE WORK

The current version is a rule-based prototype with several limitations. It depends on the user supplying wind speed and wave height, and does not measure them itself. It uses fixed thresholds adapted from [1]; official advisory criteria can differ by region, so the values should be validated against local agency (e.g., Indonesian) guidance before field use. It also handles only wind and waves, and does not consider visibility, currents, tides, or boat size. Finally, it does not validate input, so non-numeric or negative values are not rejected. Future work includes input validation, a configurable threshold file, a local-language interface, additional hazard classes, and connecting the program to low-cost sensors so that readings are entered automatically.

VI. CONCLUSION

This report presented a Marine Weather Decision System that addresses the lack of reliable, offline weather guidance for local fishermen. Using wind speed and wave height with a simple conditional algorithm in C++, the system produces an immediate and understandable sail or do-not-sail recommendation. Tests on representative and boundary inputs behaved as designed. With threshold validation and sensor integration, the prototype can be developed into a practical safety aid for coastal communities.

REFERENCES

- [1] National Weather Service, "Marine definitions," National Oceanic and Atmospheric Administration (NOAA), n.d. [Online]. Available: https://www.weather.gov/key/marine_definitions
- [2] B. Stroustrup, *The C++ Programming Language*, 4th ed. Upper Saddle River, NJ, USA: Addison-Wesley, 2013.